

Program : Diploma in Cyber forensics And Information security	
Course Code : 5289B	Course Title: IoT Security Lab
Semester : 5	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

Topic	Course Code	Course Name	Semester
General Knowledge in Computer Systems	1008	Introduction to IT Systems Lab	I
Basic Programming Knowledge		Problem Solving and Programming	II
Basic Knowledge in Database		DBMS Lab	III

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Introduction to arduino programming for problem solving	10	Applying
CO2	Arduino programs using loops, arrays, strings	8	Applying
CO3	Develop arduino programming with LED, button and switch	10	Applying
CO4	Interpret analog and digital communications with arduino	14	Applying
	Lab Test	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3					1	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	Develop arduino programming for problem solving		
M1.01	Introduction of Arduino IDE	2	Understanding
M1.02	Demonstrate data types, constants,	4	Applying
M1.03	Demonstrate variables, operators	4	Applying
CO2	Arduino programs using loops, arrays, strings		
M2.01	Demonstrate conditional statements, loops, arrays	4	Applying
M2.02	Demonstrate strings	4	Applying
	Lab Test – I	1.5	
CO3	Develop arduino programming with LED, button and switch		
M3.01	Demonstrate light an LED,	2	Applying
M3.02	Demonstrate 7-segment display.	2	Applying
M3.03	Demonstrate button	3	Applying

M3.04	Demonstrate switch	3	Applying
C04	Interpret analog and digital communications with arduino		
M4.01	Demonstrate UART communication protocol	3	Applying
M4.02	Demonstrate I2C communication protocol	2	Applying
M4.03	Demonstrate SPI communication protocol	3	Applying
M4.04	Open-ended Experiments	6	
	Lab Test – II	1.5	

****Open-ended Experiments:**

(Not for End Semester Examination but compulsory to be included in Continuous Internal Evaluation. Students can do open-ended experiments as a group of 2-3. There is no duplication in experiments between groups.)

1. Implement digital thermometer using Arduino UNO
2. Implement arduino calculator
3. Implement blinking LED using Arduino

Text / Reference

T/R	Book Title/Author
R1	Brian Evans, Beginning Arduino Programming, New York: Apress, 2011.
R2	Cornel Amariei, Arduino Development Cook Book, Birmingham: Packt Publishing Ltd., 2015
R3	Practical Internet of Things Security (Kindle Edition) by Brian Russell, Drew Van Duren 2. Securing the Internet of Things Elsevier
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Online Resources

Sl.No	Website Link
1	https://www.geeksforgeeks.org/how-to-create-a-forensic-image-with-ftk-imager/
2	https://medium.com/@tusharcool118/autopsy-tutorial-for-digital-forensics-707ea5d5994d